

TEACHING STATEMENT

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My teaching experience spans multiple formats—classroom teaching, discussion session leading, and one-on-one advising. Having taught as a teaching assistant and an instructor, I believe that an instructor's main job is to develop skills in critical and logical thinking using the subject as a means and teach students how to learn. The specific course provides one special case of learning, and all components of a course should be designed to show students the steps of learning a new topic. Lastly and equally importantly, students of all levels should be respected and viewed as scholars-in-training.

In both roles as a teaching assistant and guest instructor, I believe that modeling the learning process is critical. In Summer 2022, I served as a teaching assistant of two master's level statistics classes (ST 501 & ST 503—Fundamentals of Statistical Inference I & Fundamentals of Linear Models and Regression). My main duties were grading assignments and holding office hours. I maintained an open, engaging virtual environment, treating the students as future scholars rather than junior students. In Fall 2025, I gave two guest lectures at ST 421, Introduction to mathematical statistics I, in which I introduced how to find the distributions for functions of random variables. Trusting that students would master the skills given enough practice, I worked through problems to model the steps of learning, paused constantly to check in with them, and reviewed the key concepts at the end of the office hours or class.

Guided by the same teaching philosophy, the details of a course should also work together to enhance the learning experience. In Fall 2017, as an Adjunct Assistant Professor at Wake Forest University, I designed and taught MUS 101, Introduction to Western Music at WFU, a general education class for non-majors. I carefully considered what topics in music history to include and the appropriate depths to go into. I embedded writing assignments throughout the semester, since writing is an important part of studies in the humanities. I broke down the final paper into different stages—a proposal, an outline, an early draft, and a final draft—to guide students through the writing process. Rather than tasks that kept students busy, the assignments were designed to model the writing process and help students learn how to edit their own work.

Besides the learning process, I have always believed that critical thinking underlines all learning. In Summer 2025, I had the opportunity to put it into practice in Summer 2025, when I advised six undergraduate students from NC State on data science projects, a requirement of their summer internships at non-profit organizations. I met with them individually throughout the summer, guiding them through the project from identifying data, analysis, to deciding presentation format and visualization. None of my students were mathematics or statistics majors. Since they were unfamiliar with data analysis, I emphasized that the data was the means

to learn how to critically think about the findings, and data analysis was a form of story-telling and inherent in human nature. When the goal was clear, the individual steps of data collection, analysis, and communication became less intimidating. One student collected the numbers and types of legal cases at a legal aid organization between 2015 and 2025, demonstrating how the pandemic drastically increases the number of legal cases, especially about housing. Another student used assessment data from a childcare center to show the growth of children with and without special needs as well as the demographics of families benefiting from the organization.

During the same summer, I also advised high school students from North Carolina School of Science and Mathematics and put into practice my philosophy of helping students learn how to learn. For the Summer Research and Innovation Programs, I gave two tutorials (Introduction to R and Python) to all six students and advised two students for their group project on improving visualization for the Lake James Environmental Association. I held weekly office hours with my students, considered how to respond to the sponsor's requests, and suggested directions and resources. My goal was to enable my students to learn on their own. Therefore, for the visualization request, I suggested using Shiny App to construct a dashboard, provided example code, and walked them through the concept. With the suggested directions, my students quickly got into action. When they were stuck, I helped them debug in real-time, modeling how to test code and find bugs. Their final dashboard shows biological, habitat, and chemical data about Lake James, uses boxplots pre- and post-Hurricane Helene to demonstrate the effects of the flood, and uses scatter plots with linear regression lines to show correlations between levels of different chemicals. The advising experience was so rewarding that it was sad to say goodbye to them in the final meeting.

Through the different classes and roles, I remain convinced that logical thinking is the key to learning how to learn, and the subjects are lens through which to teach the essential skills. When the intuition is clear, technical details will fall into places and be more easily understood. I will implement these ideas in my future courses by providing the motivation and intuition of statistical methods, helping students think through the theory behind methodologies in a broad sense without being bogged down by details, and use tangible steps to cultivate students' learning skills. With my experience and deep interest in teaching, I am ready to work with students of all levels and majors at UNC.